1. LangChain, LangGraph, and LLM

What is the role of LangChain?

LangChain is a framework used to build applications with Large Language Models. It provides tools for managing prompts, integrating external tools, and connecting models to different data sources.

What is the role of LangGraph?

LangGraph is a framework for building stateful AI agents and workflows using graphs. Each node represents a task, while edges define the execution flow between nodes.

What is the role of an LLM?

A Large Language Model is an AI model trained to understand and generate natural language.

2. Basic LangGraph Concepts

What is a node?

A node is a function that performs a specific task in the graph.

What is an edge?

An edge defines the connection and execution flow between nodes.

What is START?

START represents the entry point of the graph.

What is END?

END represents the termination point of the graph.

Why is the state passed between nodes?

To share and update information throughout the workflow.

3. Router Evaluation and Improvements

Where did the router fail (if anywhere)? What would make the routing more robust (structured output, few-shot examples, enum validation)?

The router achieved 100% accuracy (6/6) on the six test cases. It correctly classified all mathematics, programming, and artificial intelligence questions.

However, the evaluation contains only six questions, so this result does not guarantee that the router will always perform correctly.

To make the routing more robust, we could use:

Structured output: Force the model to return a predefined category format.

Few-shot examples: Provide examples of questions and their correct categories to guide classification.

Enum validation: Restrict the accepted categories to mathematics, programming, and artificial_intelligence, and retry when the model returns an invalid category.

Conclusion

This lab was a good opportunity to understand how LangGraph works in practice. By building an intelligent assistant, I learned how to organize a workflow using nodes and edges, classify questions, connect different agents, and integrate tools and memory. The tests showed that the assistant works well for the examples used. Overall, this lab helped me improve my understanding of agentic AI and how to build more interactive AI applications.